ECEN 665 Project 4 Report: A Concurrent Dual-Band LNA+Mixer Front-End at 2.4GHz and 5.2GHz

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***Abstract*—A concurrent 2.4GHz and 5.2GHz dual-band LNA based on [1] and a single-balanced mixer in this report. The input matching network has been chosen to maximize S11 performance at 2.4GHz and 5.2GHz. The output load composed of two series LC tanks has also been designed to resonate concurrently at 2.4GHz and 5.2GHz.**

I. Introduction

CONCURRENT dual-band LNAs allow for simultaneous narrow-band operation at two different frequency bands. The difference when compared to a general wideband LNA is that the wideband LNA is more susceptible to out of band interferers which degrades the overall sensitivity of the system. A concurrent dual-band LNA easily facilitates the design of multi-band transceivers without having to create dedicated signal paths for each frequency band. Overall, this allows for the reduction of additional signal paths which results in lower cost, area, and power consumption.

To further process the amplified RF signals, frequency translation is required, and is achieved with a mixer. The mixer downconverts the RF signal into an intermediate frequency which is processed further. The mixer was optimized for conversion gain, noise figure and linearity.

The project report is organized as follows. Section II presents a background on concurrent dual-band LNA+Mixer designs. Section III covers the proposed solution including circuit design and simulation results. Section IV compares the results of this project to other reported works.

II. Background

The concurrent dual-band LNA concept was first introduced in [2] and served as the conceptual basis for this project. The concurrent architecture is an improvement to switching and parallel path LNA architectures. The concurrent dual-band architecture does not suffer with the switching time delay that the switching dual-band architecture has when changing between two bands. Also, the concurrent dual-band LNA has lower area and power consumption compared to a parallel architecture that utilizes multiple paths for each frequency band [2].

The essence of the concurrent architecture is that a complex input matching network and output load is added to a common-source LNA to make it operate at two frequency bands. The input matching network must match the source impedance at the two frequencies of interest. Likewise, the output load must resonate at the two frequencies of interest.

III. Proposed Solution

A. Concept

The dual-band LNA implemented is based off the design shown in [1]. The key differences of the dual-band LNA from a standard common source LNA presented in [3] is the input and output matching networks. The input matching and output matching networks need to be designed to resonate at two frequencies in this case 2.4GHz and 5.2GHz.

B. LNA Design Procedure

The design procedure for the input and output networks was based on the methodology presented in [1]. The main difference is that in this project the transistors were initially sized based on the current density that would provide the minimum noise figure. Simulation results showed that the optimal current density was $0.09\frac{mA}{\mu m}$. The current for the LNA was set to be 60% of the $10\mathrm{mW}$ at 3.5mA.These two values allowed the size of TN_0 and TN_1 to be calculated as

$$W_{TN_0} = \frac{I_D}{J_D} = \frac{3.5\mathrm{mA}}{0.09\frac{\mathrm{mA}}{\mu\mathrm{m}}} = 39\mu\mathrm{m} \quad (1)$$

After the transistor size was determined, a DC simulation was performed to get g_{m1} and C_{gs1}. Next, the input stage was designed based on the input impedance equation presented in [1]

$$Z_{in} = \frac{g_m L_S}{C_t} + j\{-\frac{1}{\omega C_t} + \omega L_t + \frac{\omega L_1}{1-\omega^2 L_1 C_0}\} \quad (2)$$

where

$$C_t = C_1 + C_{gs1} \quad (3)$$

$$L_t = L_S + L_G \quad (4)$$

The real part of (2) was matched to 50Ω with L_S chosen as $1\mathrm{nH}$ this allowed C_1 to be solved for

$$C_1 = \frac{g_{m1} L_s}{50\Omega} - C_{gs1} = \frac{(15.7\frac{\mathrm{mA}}{\mathrm{V}})(1\mathrm{nH})}{50\Omega} - 15\mathrm{fF} = 300\mathrm{fF} \quad (5)$$

Next, L_G was determined by setting the imaginary part of (2) equal to zero at resonance. The difficulty here was that there are two frequencies where the imaginary part needs to go to zero. Therefore, the average value of the two frequencies was used in the determination of L_G

$$\omega_{avg} = \frac{\omega_1 + \omega_2}{2} = \frac{2.4\mathrm{GHz} + 5.2\mathrm{GHz}}{2} = 3.8\mathrm{GHz} \quad (6)$$

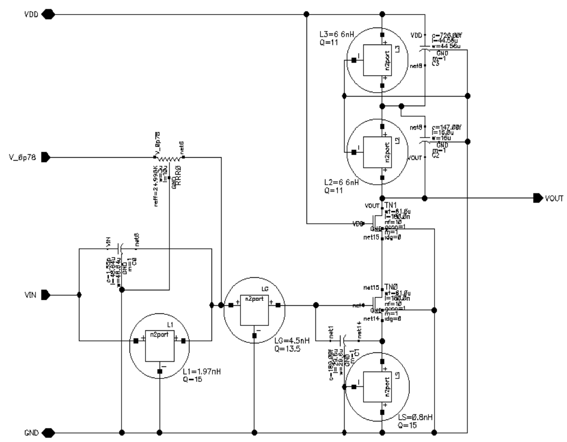

Fig. 1. Dual-Band LNA Schematic

$$L_G = L_t - L_S = \frac{1}{\omega_{avg}^2 C_t} - L_S = \frac{1}{(2\pi 3.8\text{GHz})^2(315\text{fF})} - 1\text{nH} = 4.59\text{nH} \tag{7}$$

Finally, the values of C_0 and L_1 were calculated

$$C_0 = \frac{PC_t}{SPL_tC_t - P - 1} = \frac{(2.43\text{x}10^{41})(315\text{fF})}{(1.29\text{x}10^{21})(2.43\text{x}10^{41})(5.59\text{nH})(315\text{fF}) - 2.43\text{x}10^{41} - 1} = 248\text{fF} \tag{8}$$

$$L_1 = \frac{1}{PL_tC_tC_0} = \frac{1}{(2.43\text{x}10^{41})(5.59\text{nH})(315\text{fF})(248\text{fF})} = 9.45\text{nH} \tag{9}$$

Where S and P are determined with

$$S = \omega_1^2 + \omega_2^2 = (2\pi 2.4\text{GHz})^2 + (2\pi 5.2\text{GHz})^2 = 1.29\text{x}10^{21} \tag{10}$$

$$P = \omega_1^2\omega_2^2 = (2\pi 2.4\text{GHz})^2(2\pi 5.2\text{GHz})^2 = 2.43\text{x}10^{41} \tag{11}$$

These values were placed into the schematic shown in Fig. 1. and simulated. The noise figure was too large at both frequencies so L_1 was decreased to 2nH to allow for an inductor with larger Q. Since L_1 was decreased, C_0 had to be increased to 1.55pF to maintain the input matching. To further reduce the noise figure, g_{m1} was increased by increasing W_{TN0} and W_{TN1} from 39µm to 61µm. Finally, C_1 was decreased from 300fF to 189fF to fine tune the null frequencies of the input match.

The output matching network taken from [1] consists of a series combination of two resonant tanks. Each tank was designed to resonate at each frequency band of interest.

$$\omega_1 = \frac{1}{\sqrt{L_2C_2}}, \omega_2 = \frac{1}{\sqrt{L_3C_3}} \tag{12}$$

The inductors L_2 and L_3 were chosen to be 6.3nH and the capacitors C_2 and C_3 were calculated to be

$$C_2 = \frac{1}{\omega_1^2 L_2} = \frac{1}{(2\pi 2.4\text{GHz})^2(6.3\text{nH})} = 700\text{fF} \tag{13}$$

$$C_3 = \frac{1}{\omega_2^2 L_3} = \frac{1}{(2\pi 5.2\text{GHz})^2(6.3\text{nH})} = 150\text{fF} \tag{14}$$

The final design of the LNA is presented in Fig. 1. with the addition of the 25kΩ RRR_0 for biasing purposes. Table I shows all the final dimensions for the LNA and table II describes the geometry of inductors which were simulated using SONNET.

TABLE I
TRANSISTOR DIMENSIONS (µm/µm) WITH RESISTOR VALUES, INDUCTOR VALUES, AND CAPACITOR VALUES FOR FIG. 1.

Element	Hand Calculated	Final Value
TN_0,TN_1	10(3.9/0.18)	10(6.1/0.18)
RRR_0	100kΩ	25kΩ
L_1	9.45nH	2nH
L_G	4.59nH	4.5nH
L_S	1nH	0.8nH
L_2,L_3	6.28nH	6.6nH
C_0	248fF	1.55pF
C_1	300fF	189fF
C_2	700pF	726fF
C_L	150pF	147fF

TABLE II
INDUCTOR PARAMETERS

Inductor	L_S	L_G	L_1	L_2,L_3
Shape	Spiral	Spiral	Spiral	Spiral
Turns	4	3	3	6
Metal Layer	M6	M6	M6	M6
Width (µm)	10	15	10	5
Spacing (µm)	2	2	2	2
Inner Radius (µm)	95	50	100	55
L at 2.4GHz (nH)	3.8	0.8	1.9	5.8
Q at 2.4GHz	15	14	13	12.5
L at 5.2GHz (nH)	5.1	0.8	2.1	7.4
Q at 5.2GHz	11	17	19	10

C. LNA Simulation Results

The dual-band LNA was simulated using the testbench shown in Fig. 2. where a large 1nH off-chip capacitor is used to feed the RF signal to the LNA. A bias voltage of 780mV is used to bias the LNA. Fig. 3. shows the S11, S21, and Noise Figure performance. The S11 and Voltage gain specifications were met. The NF specification was met for the 5.2GHz case, but not for the 2.4GHz case. Fig. 4. shows the IIP3 for 2.4GHz and Fig. 5. shows the IIP3 for 5.2GHz. both meet the specification. A comparison of the required LNA specifications and the achieved specifications are shown in table III.

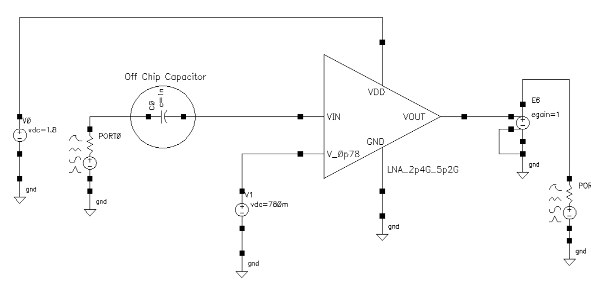

Fig. 2. Dual-Band LNA Testbench Schematic

D. Mixer Design Procedure

The mixer is implemented as a single-ended balanced mixer with PMOS current-source helper transistors. These helper transistors ease the tradeoff between the RF input transistor current and the load-resistor current. A higher input transistor current is desirable for improved linearity, whereas a lower

TABLE III
COMPARISON OF SIMULATED RESULTS AND THE REQUIRED SPECIFICATIONS FOR THE DUAL-BAND LNA

Description	Requirement	Simulated Result
NF at 2.4GHz (dB)	<2	2.98
Voltage Gain at 2.4GHz (dB)	>20	24
IIP3 at 2.4GHz (dBm)	>0	7.8
S11 at 2.4GHz (dB)	<-15	-16.8
NF at 5.2GHz (dB)	<3	2.65
Voltage Gain at 5.2GHz (dB)	>20	21
IIP3 at 5.2GHz (dBm)	>0	12
S11 at 5.2GHz (dB)	<-15	-15.6
Power Consumption (mW)	<10	8.4

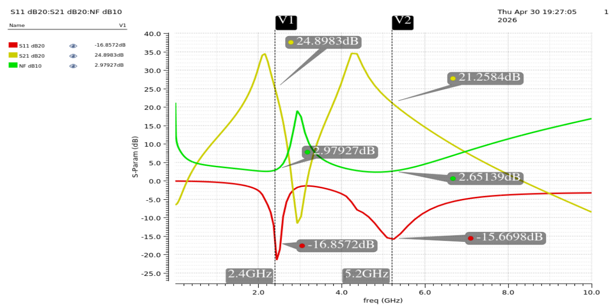

Fig. 3. Dual-Band LNA Testbench S-Parameter and NF Simulation Results

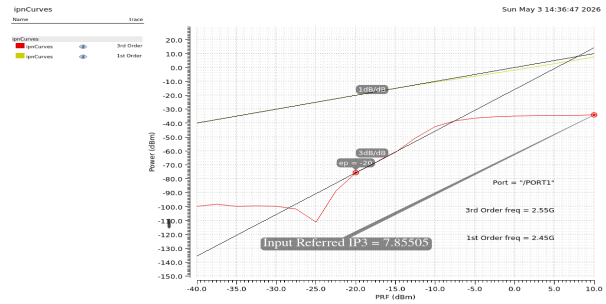

Fig. 4. Dual-Band LNA Testbench IIP3 at 2.4GHz Simulation Results

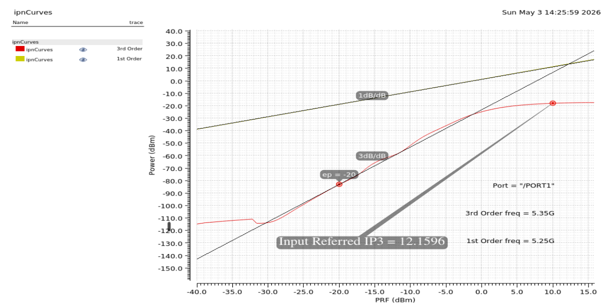

Fig. 5. Dual-Band LNA Testbench IIP3 at 5.4GHz Simulation Results

load-resistor current allows the use of a larger load resistance, thereby increasing in. By introducing current-source helper PMOS transistors in parallel with the load resistors, this tradeoff is effectively mitigated [3].

The overall mixer consists of a decoupling network and a core mixer. The core mixer includes the RF input transistor, LO switching transistors, load resistors, and PMOS current-helper transistors.

1) Decoupling Network Design: The decoupling network allows RF signals from the LNA to pass while blocking the common-mode voltage. The common-mode voltage for the mixer is generated using a current-biased diode-connected transistor, which also sets the bias current of the core mixer.

The cutoff frequency of the decoupling network is chosen such that it is at least 10 times lower than the lowest RF frequency:

$$\frac{1}{2\pi RC} < \frac{2.4\,\text{GHz}}{10} \tag{15}$$

Based on this, the selected values are:

$$R = 10\,k\Omega, \quad C = 5\,pF \tag{16}$$

2) RF Input and Switching Transistor Design: From the g_m versus I_D analysis, a transistor sizing of:

$$\left(\frac{W}{L}\right) = 20 \times \frac{2.4\mu m}{0.4\mu m} \tag{17}$$

achieves approximately:

$$g_m \approx 10\,\text{mS} \quad \text{at} \quad I_D = 2\,\text{mA} \tag{18}$$

The corresponding overdrive voltage is:

$$V_{OV} \approx 368.8\,\text{mV} \tag{19}$$

A higher V_{OV} improves linearity but reduces gain (since gain is proportional to $g_m R_{load}$) and degrades noise performance. The LO common-mode voltage is chosen as:

$$V_{LO,CM} = 2.4\,\text{V} \tag{20}$$

To ensure saturation of the RF input transistor:

$$V_{LO,BIAS} - V_{GS,\text{switch}} > V_{OV,\text{RF transistor}} \tag{21}$$

The switching transistor sizing is:

$$\left(\frac{W}{L}\right) = 20 \times \frac{4\mu m}{0.18\mu m} \tag{22}$$

3) Load Resistance and PMOS Current Helper: The value of R_{load} and the PMOS helper transistor biasing are determined by the current-splitting factor α, defined as the ratio of current flowing through the helper transistor to the total current. The PMOS helper transistor is biased near the boundary of triode and saturation to limit its noise contribution.

$$R_{load} = 0.8\,k\Omega \tag{23}$$

The conversion gain is given by:

$$A_{conv} = \frac{2}{\pi} g_m \left(R_{load} \parallel r_{ds,\text{helper}}\right) \tag{24}$$

$$A_{conv} \approx 7.01\,\text{dB} \tag{25}$$

4) LO Frequency and Swing Requirements: In a concurrent dual-band LNA system, careful selection of the LO frequency is required to avoid image overlap between the two RF bands [1]. If the LO is placed improperly, signals at 2.4 GHz and 5.2 GHz can become images of each other. To avoid this, the LO frequency is chosen as:

$$f_{LO} = 3.35\,\text{GHz} \tag{26}$$

The resulting intermediate frequencies are:

$$IF_1 = 3.35 - 2.4 = 0.95\,\text{GHz} \tag{27}$$

$$IF_2 = 5.2 - 3.35 = 1.85\,\text{GHz} \tag{28}$$

The LO swing is chosen as:

$$V_{LO,pp} = 600\,\text{mV} \tag{29}$$

A higher LO swing improves switching efficiency, which enhances gain and reduces noise.

5) Supply Voltage:

$$V_{DD} = 3.5\,\text{V} \tag{30}$$

6) Power Consumption: The total power consumption is given by:

$$P = (2\,\text{mA}) \cdot (3.5\,\text{V}) + (50\,\mu\text{A} \times 2 + 100\,\mu\text{A}) \cdot (3.5\,\text{V}) \tag{31}$$

where 100uA current is consumed in the bias for the RF transistor and 50uA in the bias for PMOS current helper transistor.

$$P \approx 7.7\,\text{mW} \tag{32}$$

7) Load Capacitor: To suppress LO feedthrough, load capacitors of:

$$C_{load} = 250\,\text{fF} \tag{33}$$

are used at the mixer output.

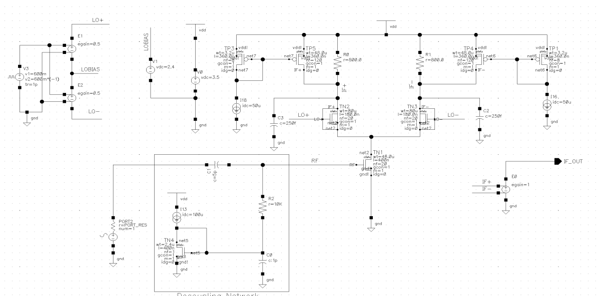

Fig. 6. Active Single-ended balanced Mixer Schematic

E. Mixer Simulation Results

The performance of the mixer was verified through conversion gain, noise figure, and linearity simulations.

1) Conversion Gain: a) RF Input at 2.4 GHz: For an input RF signal at 2.4 GHz, the resulting intermediate frequency (IF) is:

$$IF = f_{LO} - f_{RF} = 3.35 - 2.4 = 0.95\,\text{GHz} \tag{34}$$

The corresponding conversion gain plot is shown in Fig. 7.

b) RF Input at 5.2 GHz: For an input RF signal at 5.2 GHz, the IF is:

$$IF = f_{RF} - f_{LO} = 5.2 - 3.35 = 1.85\,\text{GHz} \tag{35}$$

The corresponding conversion gain plot is shown in Fig. 8.

2) Noise Figure: The single-sideband (SSB) noise figure of the mixer was evaluated at both IF frequencies:

- NF at $IF_1 = 0.95\,\text{GHz}$ is $10.76\,\text{dB}$
- NF at $IF_2 = 1.85\,\text{GHz}$ is $10.508\,\text{dB}$

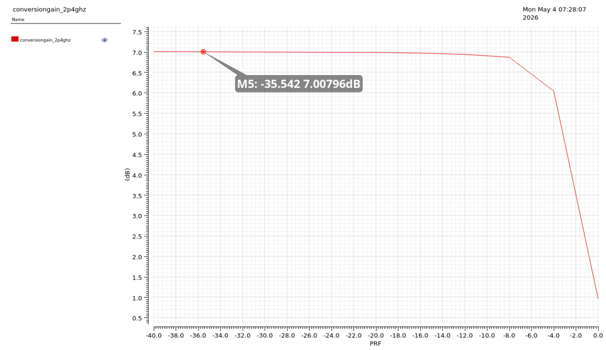

Fig. 7. Conversion Gain vs Signal Amplitude at 2.4GHz

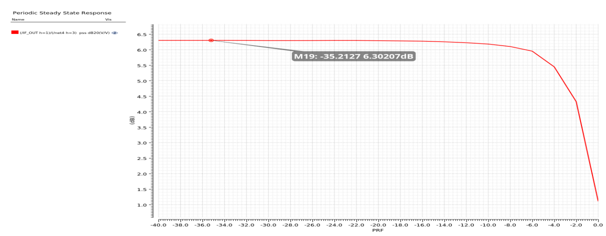

Fig. 8. Conversion Gain vs Signal Amplitude at 5.2GHz

3) IIP3 Characterization: Two-tone simulations were performed to evaluate the third-order intermodulation distortion (IIP3) of the mixer.

a) Test around 2.4 GHz band: Two tones at $F_1 = 2.35\,\text{GHz}$ and $F_2 = 2.45\,\text{GHz}$ were applied.

$$\text{IM3} = f_{LO} + F_1 - 2F_2 = 3.35 + 2.35 - 2 \times 2.45 = 0.80\,\text{GHz} \tag{36}$$

$$\text{Fundamental IF} = f_{LO} - F_2 = 3.35 - 2.45 = 0.90\,\text{GHz} \tag{37}$$

The corresponding IIP3 is 9.667dBm and plot is shown in Fig. 10.

b) Test around 5.2 GHz band: Two tones at $F_1 = 5.15\,\text{GHz}$ and $F_2 = 5.25\,\text{GHz}$ were applied.

$$\text{IM3} = 2F_1 - F_2 - f_{LO} = 2 \times 5.15 - 5.25 - 3.35 = 1.70\,\text{GHz} \tag{38}$$

$$\text{Fundamental IF} = F_1 - f_{LO} = 5.15 - 3.35 = 1.80\,\text{GHz} \tag{39}$$

The corresponding IIP3 is 6.25dBm and its plot is shown in Fig. 11.

IV. Discussion of Results

A comparison of concurrent dual-band LNAs is shown in table V. This work has the largest gain and IIP3 of the works compared. However, these simulated results are highly optimistic compared to the measured results of the other works. The designed mixer achieves a conversion gain greater than 6 dB for both IF frequencies, with a SSB noise figure close to 10 dB and an IIP3 of approximately 10 dBm for the fundamental frequency of 2.4GHz, meeting the intended design targets. However, like the LNA these results are simulated and therefore optimistic.

The main area for improvement in this LNA+Mixer design is in the NF performance at 2.4GHz. This project was unable

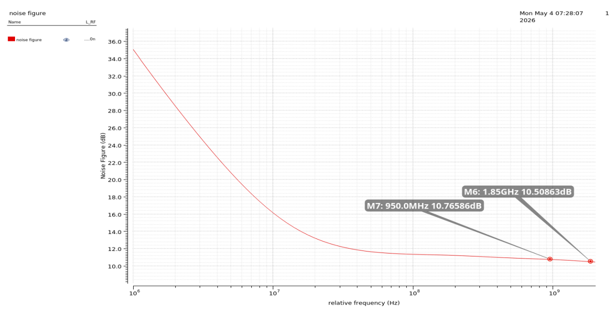

Fig. 9. Noise Figure Plot

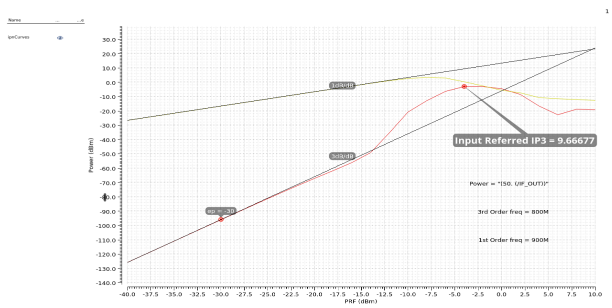

Fig. 10. IIP3 characterization for 2.4 GHz band

to meet the required specification of <2dB. One possible solution is to employ higher Q inductors at the input matching network as this would reduce the parasitic resistance and lower the NF.

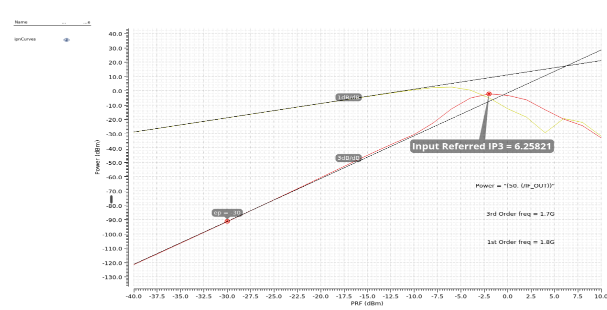

Fig. 11. IIP3 characterization for 5.2 GHz band

TABLE IV
COMPARISON OF SIMULATED RESULTS AND THE REQUIRED SPECIFICATIONS FOR THE MIXER

Description	Requirement	Simulated Result
NF at IF1 (dB)	<10	10.76
Voltage Gain at 2.4GHz (dB)	>6	7.007
IIP3 at 2.4GHz (dBm)	>10	9.667
NF at IF2 (dB)	<10	10.508
Voltage Gain at 5.2GHz (dB)	>6	6.3
IIP3 at 5.2GHz (dBm)	>10	6.25
Power Consumption (mW)	<10	7.7

TABLE V
COMPARISON OF CONCURRENT DUAL-BAND LNAS IN THE 2.4GHZ AND 5.2GHZ BANDS

Ref.	Technology (μm)	Freq. (GHz)	S11 (dB)	S21 (dB)	NF (dB)	IIP3 (dBm)	Power (mW)
[2]	0.35 CMOS	2.4 5.2	-25 -15	9.4 15.5	2.3 4.5	0 5.6	10
[4]	0.18 CMOS	2.6 5.2	-20 -22	15 9.5	0.6 0.8	0 7	3.6
[5]	0.18 CMOS	2.4 5.2	-15 -18	14 22	2.4 2.5	-0.2 -6.3	9.3
[6]	0.18 CMOS	2.4 5.2	-13.1 -10.5	12.9 8.2	3.7 3.7	-4 -1	7.6
[1]	0.13 CMOS	2.4 5.2	-16.8 -19.4	19.3 17.5	3.2 3.3	-20.1 -18.1	2.4
This Project	0.18 CMOS	2.4 5.2	-16.8 -15.6	24 21	2.97 2.65	7.8 12	16.1

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